Coordinate Geometry

The big idea

How do you describe exactly where a point is on a flat surface? René Descartes used two perpendicular number lines. Every point then gets a unique pair of numbers, its **coordinates**.

The Cartesian plane

- The horizontal line is the **x-axis**, the vertical line the **y-axis**; they meet at the **origin O (0, 0)**.
- A point P is written **(x, y)**: x is the **abscissa** (distance from the y-axis), y is the **ordinate** (distance from the x-axis).
- The axes divide the plane into four **quadrants**.

Quadrant	Signs	Example
I	(+, +)	(3, 2)
II	(−, +)	(−3, 2)
III	(−, −)	(−3, −2)
IV	(+, −)	(3, −2)

POINTS ON THE AXES

A point on the **x-axis** has y = 0, like (5, 0). A point on the **y-axis** has x = 0, like (0, −4). Points on the axes don't belong to any quadrant.

IN WHICH QUADRANT OR ON WHICH AXIS ARE THESE POINTS?

(−2, 4): quadrant II. (3, −1): quadrant IV. (−1, 0): on the negative x-axis. (0, 0): the origin.

(X, Y) IS AN ORDERED PAIR

(3, 5) and (5, 3) are different points. The x-coordinate always comes first.

Plotting points

To plot (3, −2): start at the origin, move 3 units **right** along the x-axis, then 2 units **down**. Mark the point.

Quick recap

- Two perpendicular axes meet at the origin (0, 0).
- (x, y): abscissa first, ordinate second.
- Signs in quadrants: I (+, +), II (−, +), III (−, −), IV (+, −).

Practice questions

1. The point (−3, 5) lies in quadrant
   (a) I (b) II (c) III (d) IV
2. The ordinate of the point (4, −7) is
   (a) 4 (b) −7 (c) 7 (d) −4
3. A point on the y-axis has
   (a) x-coordinate 0 (b) y-coordinate 0 (c) both coordinates equal (d) positive coordinates
4. The coordinates of the origin are
   (a) (1, 1) (b) (0, 0) (c) (0, 1) (d) (1, 0)
5. Points with both coordinates negative lie in quadrant
   (a) I (b) II (c) III (d) IV

Answers: 1. b 2. b 3. a 4. b 5. c